# Exercise 5: A four-of-a-kind game

A standard pack of playing cards has four suits: Hearts, Spades, Diamonds and Clubs. In each suit, there are 13 ranks: 9 numbered cards (2-10), and 4 picture cards (Ace, Jack, Queen, King). Let's make a game where a player attempts to draw four-of-a-kind (i.e. four cards of the same rank) in a certain number of draws from the deck. For example, given 10 draws, it might be that the player draws three 4s (of any suit) amongst other cards, but not a fourth 4, and so they will not win.

**1.** Add the following lines of Python to begin with:

```
suits = ( "H", "S", "D", "C" )

ranks = tuple( range( 1, 14 ) )

cards = ()
```

Let's represent each card as a string. For example, the Ace of Hearts might be "H1", and the Eight of Diamonds might be "D8".

Now, write a function "add_cards" which takes a single suit string (H, S, D or C) as an argument, and adds all the ranks of that suit to the cards tuple. Remember: "cards" cannot be written to from within a function unless it is considered global, and a tuple cannot be added to, only a whole new tuple created. (We should have used a list here, really; if you find this part of the exercise too difficult, feel free to change "cards" to be a list instead).

**2.** Add code which iterates over all the suits, and calls your "add_cards" function to build a list of all 52 cards in a pack.

**3.** Add code which will make a copy of the "cards" tuple, but as a _list_ (which we can then alter, if we need to). Call that variable "deck". Add code to create a whole new variable called "hand" (to represent the cards the player is holding), and make it an empty list.

**4.** Next, write a function called "take_card", which picks a card at random from the "deck" list, adds it to the "hand" list, and removes it from the "deck" list. Remember, you can import the "random" module, and use "random.choice( <your_list> )" to get a random element from your list.

**5.** Make a new variable called number_of_draws, and set its value to 10 (for now). Add a for loop which will call your "take_card" function "number_of_draws" times, and print out "deck" and "cards" at the end of it to make sure the cards are moving correctly from the deck to the hand.

**6.** Finally, make a function "check_win" which returns True or False, after checking to see whether there are four cards of the same rank (ignoring the suit) in the player's "hand" list. If this seems difficult to do, perhaps consider using a dictionary whose keys are the rank strings ("1" to "13"), and whose values are an integer, to keep track of how many of each rank are present. Have your code call your function, and print out a message to let the player know if they won or not.

What number of draws do you notice are needed to ensure that the player wins the game most of the time, from running your code many times with different values for "number_of_draws"?